

THE UNLOX ACADEMY

Unsupervised ML Minor Project - Project Spec & Data Dictionary

QuickCart Market Basket Analysis

Apriori & FP-Growth on 5,000 Real Baskets

QuickCart (same brand as the K-Means clustering project in Week 8 and the Stockout Risk classification project) wants to know: which products get bought TOGETHER? Not which customers look alike, not which SKU will run out - which items belong next to each other on a shelf, in a bundle deal, or in a 'frequently bought together' recommendation.

The technique behind every 'customers who bought X also bought Y' feature you've ever seen. It's a genuinely different unsupervised technique from the K-Means clustering already covered in Week 8, and it's specifically the technique students ask about most and get the least hands-on practice with.

DESIGN NOTE**Why this dataset?**

The 'Groceries dataset' (9,835 transactions) on Kaggle is THE most tutorialized Apriori dataset in existence - a working notebook is one search away for any student or LLM. This dataset is custom-built, unpublished, and reuses the exact 60-SKU QuickCart catalog students already know from Weeks 8 and 9, so results can be verified but not looked up.

It also lets us plant KNOWN, VERIFIABLE association patterns at controlled strength - impossible to do with a found dataset - so every mined rule below has a locked, correct answer.

1. The Business Problem

5,000 QuickCart orders across 12 stores over November 2026, drawn from the same 60-SKU catalog used in the Stockout Risk project. Task: mine ASSOCIATION RULES - patterns of the form 'if a basket contains A, it's likely to also contain B' - and translate the strongest rules into real business actions: shelf placement, bundle deals, and cross-sell recommendations.

The three core metrics

Metric	Formula	What it answers
Support	$P(A \text{ and } B) = \text{orders with both} / \text{total orders}$	How common is this combination overall?
Confidence	$P(B A) = \text{orders with both} / \text{orders with A}$	Given someone bought A, how likely are they to also buy B?
Lift	$\text{Confidence} / P(B)$	Is this combination more common than random chance? (>1 = yes, meaningfully)

Lift is the metric that matters most for filtering real signal from noise. A lift of 1.0 means A and B are statistically independent - no relationship, buying A tells you nothing about B. A lift of 5.0 means baskets with A are 5x more likely to also have B than random chance would predict.

2. Schema - 1 Table, Long Format

Deliberately simple schema - the complexity in this project lives in the ALGORITHM (Apriori/FP-Growth), not in multi-table joins like the Stockout Risk project.

quickcart_basket_transactions.csv (26,878 rows)

One row per (order, SKU) line item - a 'long' format transaction log, the standard shape for basket analysis.

Column	Type	Notes
order_id	string	ORD-00001 to ORD-05000 - 5,000 unique baskets
date	date	Nov 1–30, 2026
store_id	string	ST01–ST12 - same 12 QuickCart stores as the Stockout project
sku_id	string	SKU001–SKU060 - same catalog as the Stockout project
sku_name	string	Product name (denormalized for convenience)
category	string	8 categories (denormalized for convenience)

TEACHING TIP

The first real step is a RESHAPE, not a clean.

This CSV is in LONG format (one row per item). Apriori/FP-Growth need a basket-level view - either a list of items per order_id, or a one-hot encoded WIDE matrix (one row per order, one column per SKU, True/False). Converting long → basket-list → one-hot is the first real task, and it's a genuinely different data-shaping skill from anything in the Stockout Risk project.

3. Locked Sanity Numbers

Row counts:

quickcart_basket_transactions.csv:	26,878 line items
Unique orders (baskets):	5,000
Unique SKUs appearing:	60 of 60 (full catalog used)
Date range:	2026-11-01 to 2026-11-30
Stores represented:	12

Basket size distribution:

Mean:	5.38 items/basket
Median:	5
Min/Max:	1 / 16

Top 5 single-item support (most purchased SKUs):

White Bread 400g	25.46%
Toned Milk 1L	25.40%
Potato Chips 90g	21.84%
Cola 750ml	21.64%
Toor Dal 1kg	20.68%

Frequent itemsets found (min_support = 0.02, via mlxtend apriori):

1-itemsets:	60
2-itemsets:	78
3-itemsets:	86
4-itemsets:	30
5-itemsets:	2
Total:	256

Association rules generated (min_lift threshold = 1.0): 980

4. The 6 Planted Patterns - Locked Answer Key

Six deliberate co-purchase patterns were planted at varying strength. Every one surfaces as a clean, high-lift, high-confidence rule - verify your own Apriori run against these numbers.

Pattern	Rule (A → B)	Support	Confidence	Lift
Breakfast Combo	Toned Milk → White Bread	23.84%	93.86%	3.69
Thali Staples	Basmati Rice → Toor Dal	15.34%	77.40%	3.74
Snack-Drink Combo	Potato Chips → Cola	19.74%	90.38%	4.18
Ghar Breakfast	Ghee → Poha	14.68%	86.05%	5.07
Tea-Time	Instant Coffee → Choco Cookies	15.20%	85.88%	4.84
Personal Care Restock	Hand Sanitizer → Face Wash	13.18%	79.30%	5.05

GOTCHA

A genuinely interesting caveat that showed up unplanned: compound high-lift rules.

The single highest-lift rules in this dataset (lift up to ~26.8) are NOT the 6 planted pairs above - they're 2-item-to-2-item rules formed when TWO planted patterns happen to co-occur in the same basket (e.g. a basket containing both the Tea-Time pattern AND the Personal Care Restock pattern). Their support is much lower (~2%) than the core pairs.

This is a genuine, valuable teaching moment: very high lift on very low support is a common trap. A rule with lift=26.8 sounds more impressive than lift=5.0, but with only ~100 supporting baskets out of 5,000, it's far less reliable and far less actionable than a strong core pair with 15–24% support. Teach students to look at BOTH numbers together, never lift alone.

The control check - proof the patterns are real, not noise

A random, unplanted pair (Banana 1kg → Shampoo 340ml - two categories with no designed relationship) did not even clear the 2% minimum support threshold to appear in the rules table at all. This is worth demonstrating explicitly: it's the clearest possible proof that the 6 planted patterns are genuine signal, not an artifact of the mining method finding patterns in anything.

5. Suggested Analysis Path

Step 1 - Reshape to basket format

```
baskets = tx.groupby('order_id')['sku_name'].apply(list).tolist()

from mlxtend.preprocessing import TransactionEncoder
te = TransactionEncoder()
te_ary = te.fit(baskets).transform(baskets)
df_encoded = pd.DataFrame(te_ary, columns=te.columns_)
```

Step 2 - Mine frequent itemsets

```
from mlxtend.frequent_patterns import apriori, fpgrowth

# Apriori: intuitive, slower on wide catalogs
frequent_itemsets = apriori(df_encoded, min_support=0.02, use_colnames=True)

# FP-Growth: same result, much faster on larger catalogs -
# worth benchmarking BOTH and comparing runtime
frequent_itemsets_fp = fpgrowth(df_encoded, min_support=0.02, use_colnames=True)
```

Step 3 - Generate and filter rules

```
from mlxtend.frequent_patterns import association_rules

rules = association_rules(frequent_itemsets, metric='lift', min_threshold=1.0)
rules = rules.sort_values('lift', ascending=False)

# Filter to ACTIONABLE rules: reasonable support AND strong lift
actionable = rules[(rules['support'] > 0.05) & (rules['lift'] > 2)]
```

Step 4 - Translate rules into business recommendations

- Shelf placement: which SKUs should be physically near each other in a dark-store picking layout?
- Bundle deals: which pairs are strong enough to package as a discounted combo?
- App recommendations: 'frequently bought together' prompts at checkout - which rules have both high confidence AND reasonable support?

6. File Manifest

File	Rows	Purpose
dim_skus.csv	60	SKU catalog (same as Stockout Risk project)
quickcart_basket_transactions.csv	26,878	The core dataset - long-format transaction log
frequent_itemsets.csv	256	Reference/answer-key - pre-computed frequent itemsets (mentor use)
association_rules_full.csv	980	Reference/answer-key - all mined rules (mentor use)

Continuity note: QuickCart is the same brand and 60-SKU catalog used in Week 8 (K-Means clustering) and the Stockout Risk classification project. Students will recognize every product name. No shared keys with either prior dataset - this is an independent transaction log.